

General Industries Hub

Enterprise Inventory Management

HOD Manual

Head of Department oversight: approvals, EOD, reports.

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GI Hub - Role booklet - HOD Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no
Two-stage approval	Store Keepers stage entries step-HOD ready and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal
Audit-first	Every action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.
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1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock	Permanent - stock figures move now
4	System	Sends Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then records the physical arrival from	Warehouse - Receiving
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals

Step	Role	Action	Lands in
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on	Permanent stock record; the

Three side-paths exist for when things do not go to plan: site delivery closes

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
2	Head of Department	One site
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Admin is the single exception and reaches every workspace deliberately, for support.

2.2 Page access matrix

Page	Store Keeper	Warehouse User	Supervisor	HOD	Logistics	Auditor	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Entry Log	[OK]	[X]	[X]	[X]	[X]	[X]	[X]
HOD Portal	[X]	[X]	[X]	[OK]	[X]	[X]	[X] (hidden - uses Admin)
Logistics Portal	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (admin)
Warehouse Portal (NEW)	[X]	[OK]	[X]	[X]	[X]	[X]	[OK] (admin shadow, picks WH)
Admin	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]	[OK]	[OK] (read + download only)	[OK]

Records (Inventory, ledgers, POs, PRs)	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[OK] (admin
Material Estimator	[X]	[X]	[X]	[OK]	[X]	[X]	flow) [OK]
Man-Hours (NEW)	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Auditor view	[X]	[X]	[X]	[X]	[X]	[OK]	[OK]

> Exact-role locks: Entry Log, HOD Portal, Supervisor Portal, Logistics Portal, Warehouse Portal, Material Estimator and Man-Hours are locked to their own role - a higher-ranked role does not inherit access. Admin reaches them for support purposes.

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User (NEW)	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
HOD	Their own site only - but they can REQUEST material from
Logistics (NEW)	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Auditor (NEW)	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar

when

shadowing the Warehouse Portal.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD

Username	Password	Role
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics or warehouse_user accounts are seeded. Both roles are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session

Element	Purpose	Notes
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

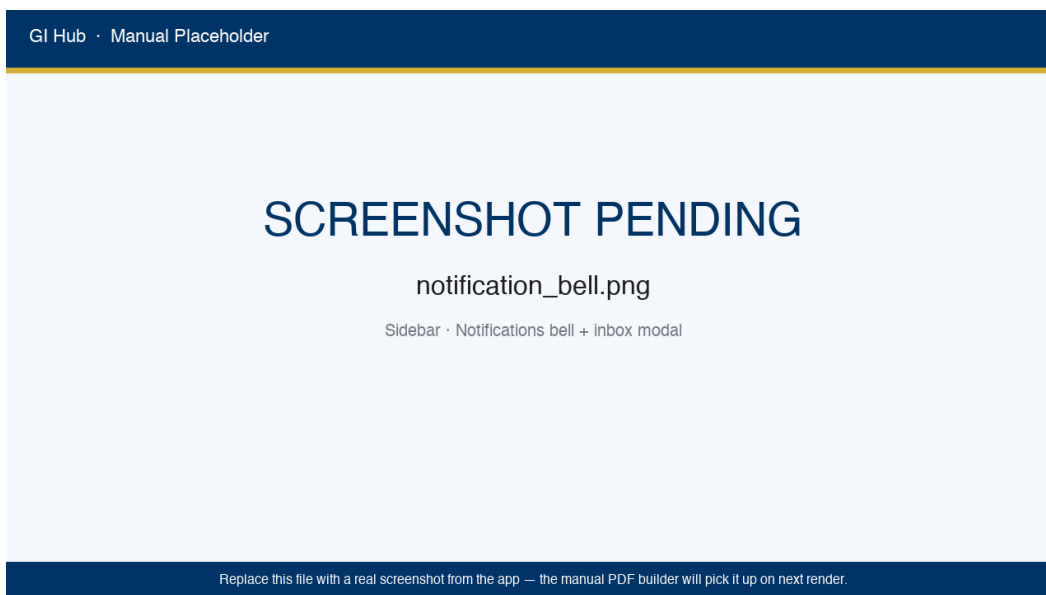
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

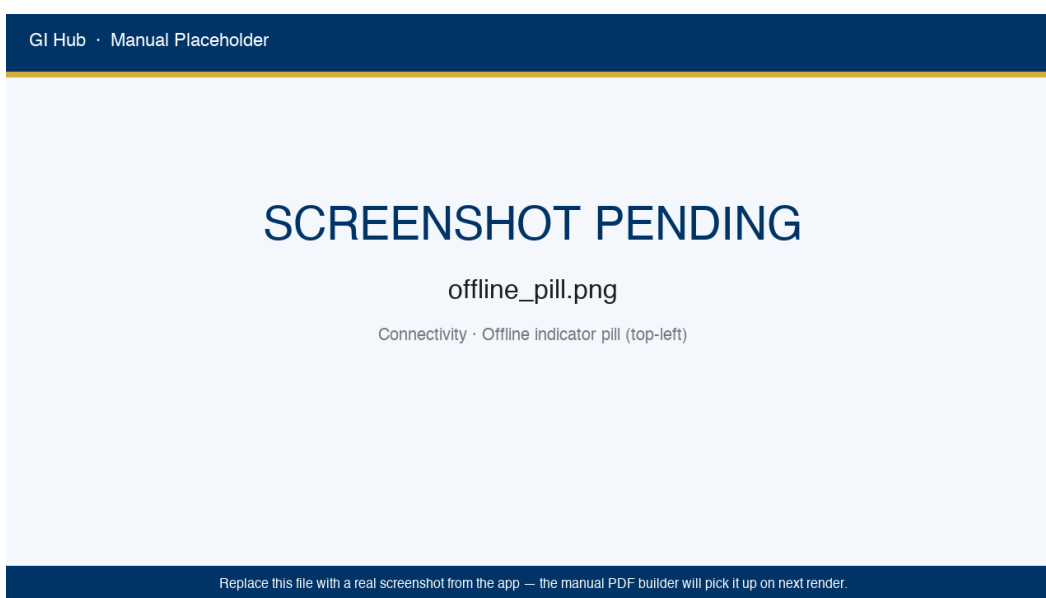
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope)

you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if a notification helper errors, the badge silently shows 0 instead of crashing the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by toggles in config.WHATSAPP_TRIGGERS (see §9).

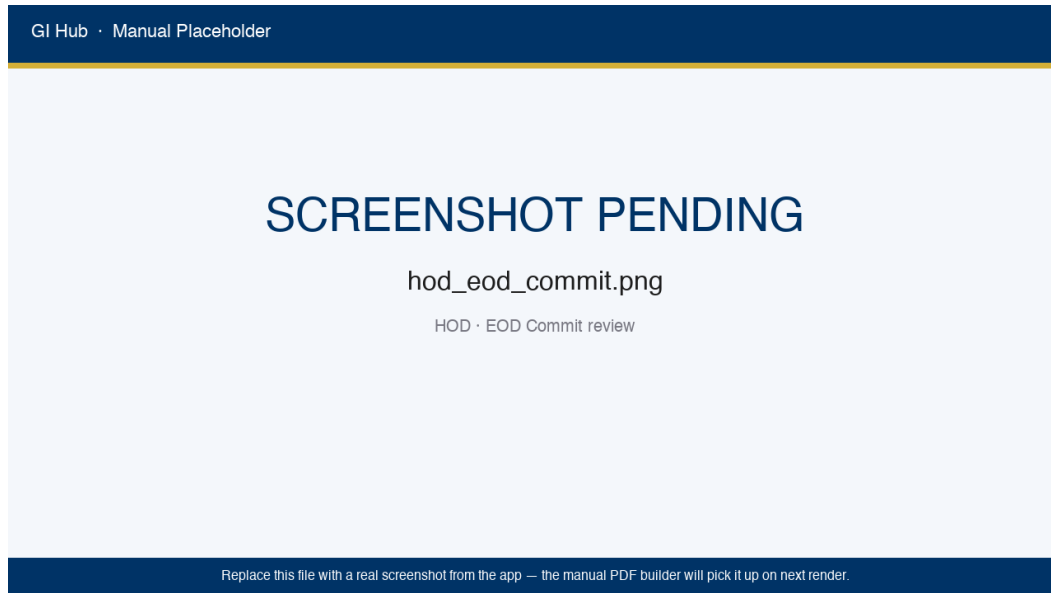
3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

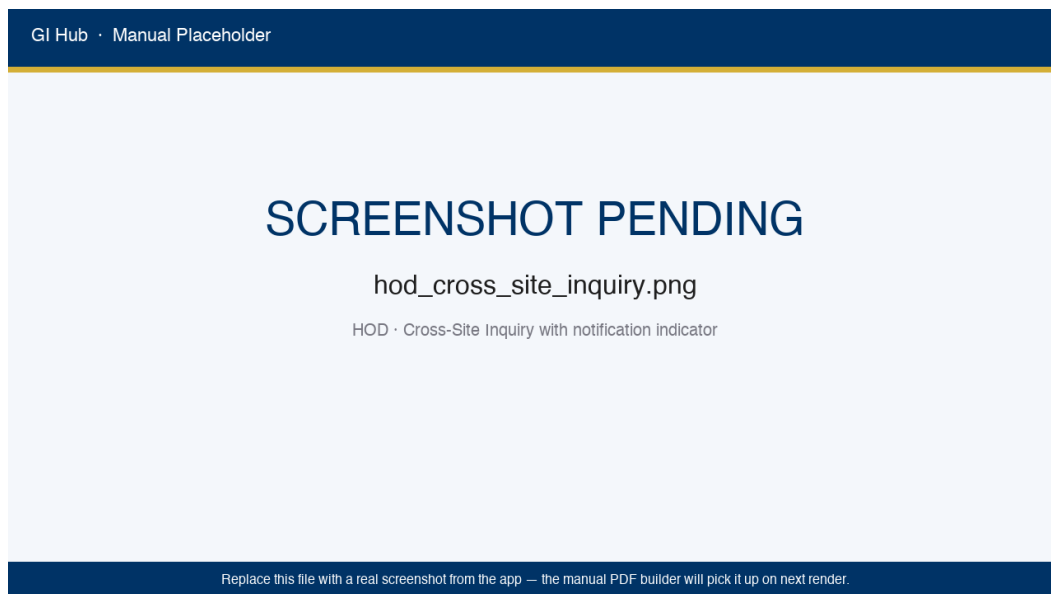
> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

6. HOD (Head of Department) Manual

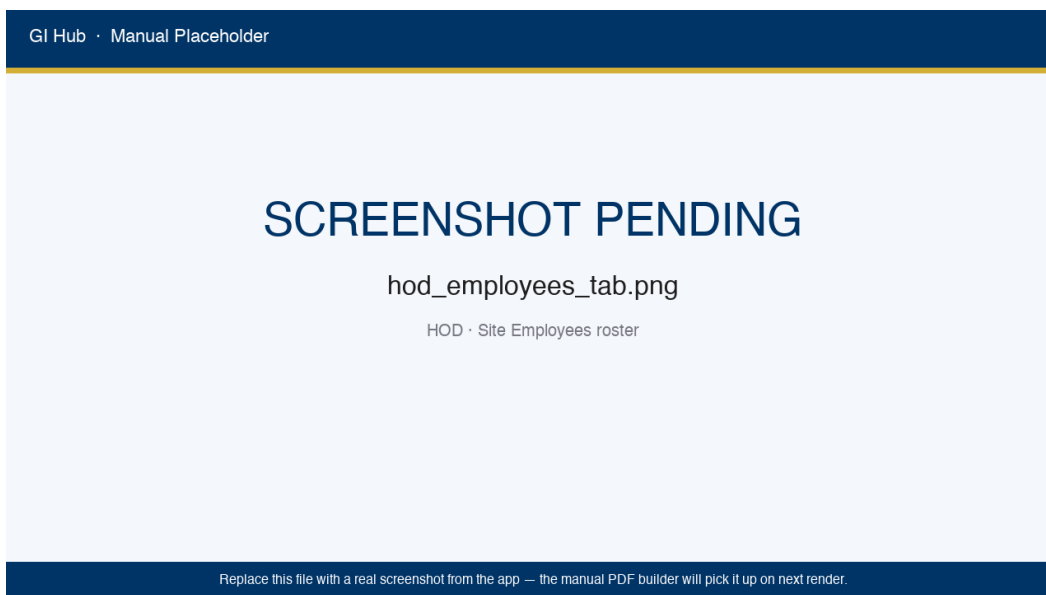
The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has 16 tabs as of v3.0 - covering EOD, Cross-Site, Burn Rate, Receipts, Returns, Adjustments, PRs, Shelf-Life, Notifications, My Requests, Site Config, the new Employees tab (Phase 7A), DOC, QR Approval, DN Approvals, In-Transit.



HOD Portal - End-of-Day Commit review with negative-stock guard



HOD Portal - Cross-Site Inquiry with the Phase 7C "you are viewing" notification banner

*HOD Portal - Site Employees roster with add/edit form (Phase 7A)*

6.1 Pages visible

- Live Dashboard (see §5.2)
- Entry Log (see §4 - HODs can also stage entries)
- HOD Portal - detailed below
- Reports (see §8)

6.2 HOD Portal overview

6.2.1 Page header

render_brand_header_hod("HOD Management Portal") - purple subtitle + today's date.

- Page title: " HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	cached_total_inventory_value(site_id)	Delta: <SAR> consumed (30d) - site-scoped
Below minimum (site)	cached_low_stock_items(site_id) count	Green/amber/red
Expiring / expired	cached_short_dated_stock(site_id) count	Green/amber/red
Pending receipts to approve	Count from pending_receipts where status='pending_hod' at your site	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit

2.

Cross-Site

3. Burn Rate
4. Pending Receipts
5. Adjustments
6. Purchase Requests
7. Receive Material
8. [!] Shelf-Life
9. Notifications
10. [OK] My Requests

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Sets every pending row to status='approved'. Disabled when count = 0.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing every column from the staging row (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects Lot_Number and FEFO_Override columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row (sets status='approved')
[X] button	Reject this single row (sets status='rejected')

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click Commit EOD to Master:

Step 1 - pre-flight: negative-stock guard

Before committing, the system checks that nothing would push stock below zero. If anything would:

Element	Purpose
Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume -
Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort
Confirm Commit button (red)	moves rows to consumption, deletes from pending_issues, busts caches, queues a post-EOD low-stock alert to the HOD if applicable.

A confirmation appears once the commit succeeds.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	min(requested, available)
Add to List button	Adds to the cart

6.4.3 Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes

Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site
Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests addressed to their site from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Calls update_request_status(status='approved') + audit
Rejection action	Same with status='rejected'

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)
- Per-row line: material name (truncated) - gradient bar (color reflects burn intensity proportional to max) - monthly figure with UOM - days-remaining badge (red if <=7d, amber if <=30d, green if >30d)

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed forecast table

Columns: SAP_Code - Equipment_Description - UOM - Current_Stock - Daily_Burn_Rate - Days_Remaining

Sortable + exportable.

6.6 HOD Portal -> Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: " N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Bulk: calls `commit_pending_receipts(site_id, username)` which:

1. Iterates every pending receipt
2. Calls `process_receipt_delivery` (which creates lots if `Expiry_Date` set)
3. Posts to receipts table
4. Auto-closes PR if fulfilled
5. Deletes the staged rows
6. Busts caches, audit-logs, success toast

6.6.3 Receipt table

Columns: Date - SAP - Equipment_Description - UOM - Quantity - Supplier - PR_Number - Site_ID - Status pill

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier
[X] Reject button	Soft-reject (row stays as status='rejected'), preserves audit, audit-logs with reason

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
Pending	Count of pending adjustments at your site
Shortfalls	Variance < 0 count
Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - Variance (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Calls approve_stock_adjustment(adj_id, approver). Atomically posts a synthetic ledger row - if variance < 0, inserts into consumption with Work_Type=STOCK_ADJUSTMENT; if > 0, inserts into receipts with Supplier=STOCK_ADJUSTMENT. The posted_txn_ref ('C:rowid' or 'R:rowid') links back to the adjustment for full audit.
[X] button	Soft-reject. Row stays as status='rejected' for audit.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - Status pill - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Calls insert_manual_pr(...). Sets status='open', workflow_state='draft'

6.8.2 Upload PR PDF (auto-extract via pdfplumber) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	extracts PR number + Material Codes (GI-XXXXXXX pattern), matches to inventory's Material_Code, inserts pr_master rows
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - Workflow pill

Pending_Qty = Requested_Qty - SUM(receipts where PR_Number matches)

6.8.4a Submit PR(s) to Logistics Portal (NEW in v3.0)

A new expander sits between the PR list and the email/PDF block. It opens the procurement-chain path - handing off your PR to the in-app Logistics queue instead of (or in addition to) the legacy email path.

Element	Purpose
Multi-select	All open PRs at your site that haven't yet been submitted to
Submit Selected to Logistics button	The <code>logistics_status</code> flips to submitted and the row appears in Logistics Portal -> Incoming PRs. A <code>pr_submitted_to_logistics</code> notification fires to the Logistics role inbox + WhatsApp (if enabled).

The legacy email path (§6.8.4) still works - your team can use it for direct-to-vendor relationships that don't go through central Logistics. Both paths coexist. The email path is marked for future deprecation once procurement chain adoption reaches the agreed threshold.

6.8.4 Notify Logistics section

When at least one PR has status='open':

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table.
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts (Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	instantly posts to receipts AND creates a lots master row if Expiry_Date is set.	

Auto-closes PR if fulfilled. Busts

6.9.3 Receipt History (last 50) caches.

Below the form, a table showing recent receipts for this site: rowid - Date - SAP - Material - Quantity - Supplier - PR_Number - Expiry_Date.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
Critical (<=30d)	Days_left 0-30
Warning (<=90d)	Days_left 31-90

6.10.2 Action-required banner

When expired_n > 0: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - Days Left - Status pill

6.10.4 Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason expired_disposal.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Queues to whatsapp_queue with status='pending'; the background worker picks it up. Audit-logs MANUAL_WHATSAPP.

6.11.2 Alert Thresholds card

3 sliders writing to app_settings:

Slider	Range	Default	Stored key
Low stock alert (days of supply)	1-60	5	low_stock_days
Burn-rate warning (days remaining)	1-60	7	burn_alert_days
Expiry warning (days before)	1-120	30	expiry_warn_days

Save Thresholds button - persists + audit-logs UPDATE_THRESHOLDS.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - Status pill - Created

6.12.2 Mark Incoming Transfers as Received

When at least one request has status='approved':

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Calls update_request_status(status='fulfilled') + busts caches. This is where you confirm the transfer physically arrived.

6.15 HOD Portal -> DN Approvals (NEW in v3.0)

This is your approval queue for Delivery Notes inbound to your site. The Warehouse has prepared them, Logistics has approved the delivery date - now you confirm the content (what's actually arriving, in what

qty).

6.15.1 Empty state

If nothing's pending: an empty-state card reads "No DNs awaiting your approval - They appear here after Logistics signs off the delivery date." No action needed.

6.15.2 Per-DN cards

Each pending DN renders as its own bordered container with two columns:

Left column:

- Header line: DN <number> - PO <number> - From Warehouse <id>
- Subline: DN Date - Vehicle No - Driver Name
- View lines expander: Read-only preview of every line item (Material_Code, Description, Qty, UOM, rl_bl_family, Lot_Number, Expiry_Date, Remarks)

Right column (Decide popover):

- Notes textbox (required if rejecting)
- [OK] Approve button (primary) - flips DN to pending_sk and mirrors lines into pending_receipts so SK sees them
- [X] Reject button - flips DN to rejected with your reason; Warehouse gets pinged to redo

6.15.3 When to approve

- DN qty matches what your site needs (matches the PR line you originally raised)
- RL/BL family is correct
- Lot + expiry are acceptable (not over-expiry)
- The originating PO is the right one (not a misroute)

6.15.4 When to reject

- Wrong qty (e.g. site asked for 100, DN says 200 - too much exposure)
- Wrong material (the Material_Code doesn't match the PR you raised)
- Lot has insufficient remaining shelf life
- Vehicle / driver details suggest a routing problem

Always include a clear rejection reason. The Warehouse user sees the reason verbatim in their bell inbox and on the bounced DN. Sloppy reasons cost a rebuild cycle.

6.16 HOD Portal -> In-Transit (NEW in v3.0)

A read-only window onto the procurement chain for your site. Use this tab when a user asks "when is X arriving?" or "why didn't Y arrive yesterday?". Three sub-tabs.

6.16.1 Sub-tab: Active in-transit

KPI strip at the top with counts per pipeline state:

- At Logistics - DN drafted by Warehouse, waiting on Logistics date approval
- Logistics approved - date confirmed, waiting on YOU
- Awaiting my approval - same as Logistics approved (highlighted gold)
- Pending SK receipt - you've approved, SK has it in their tab

Below the KPI strip, each in-transit DN renders as a card:

Element	Purpose
DN header	DN <number> (gold) + RL/BL chip (orange for RL, purple for BL)
Subline	PO <number> - Warehouse <id> - ETA <date> - <line count> line(s), <total qty> units
Status pill (right)	Colour-coded pipeline state
View lines expander	Read-only line preview
Request reschedule popover	Date picker (defaults to ETA + 3 days, min=today) + reason

The reschedule UI is deliberately frictionless: textarea + Submit button. Submits to Logistics.

- Date is pre-filled so a quick "+3 days" submission is one click
- min_value=today prevents accidentally picking yesterday
- Same-date submission warns instead of wasting a round-trip
- Caption " Goes to Logistics" makes the destination unambiguous

6.16.2 Sub-tab: My reschedule requests

KPI strip: Pending / Approved / Rejected counts.

Custom table showing your full reschedule history for THIS site:

Column	Source
PO No.	the linked PO
DN No.	the linked DN (if any)
From	current_date when you raised the request
Requested	the new date you asked for (gold, monospace, with arrow)
Reason	your justification (tooltip-truncated)
Status	pill - pending / approved / rejected
Decided by	Logistics user who handled it
Notes	decision notes from Logistics

6.16.3 Sub-tab: Force-closures affecting me

Read-only audit table showing every PR / PO / line that Logistics force-closed, scoped to your site. Three-way fallback join means closures on records with NULL Site_ID still appear correctly via their PR or PO parent.

Column	Source
Type	"PR closed" / "PO closed" / "Line closed" badge
Target	The closed ref (PR number / PO number / line id)
PR No.	Linked PR (if applicable)
PO No.	Linked PO (if applicable)
Reason	Logistics' force-close reason
Closed by	Logistics user who closed it
When	Timestamp

50 most recent shown.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit
2. Review the 4-card stat strip
3. Sort/filter using the filter pills
4. For each row: read the full detail card (line 1 = essentials, line 2 = all fields)
5. Click [OK] to approve or [X] to reject individual rows
6. Optionally click [OK] Approve All Pending for a clean batch
7. Click Commit EOD to Master
8. If pre-flight blocks: read the violation table, close, fix the offending rows (reduce qty, drop, or receive stock first), then re-open
9. If pre-flight passes: type COMMIT exactly, click Confirm Commit
10. Watch the balloons.

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments
2. Read the pending table - variance and reason are color-coded
3. For each adjustment, decide: did the Store Keeper count accurately?
4. [OK] Approve -> system posts a synthetic ledger row (receipt or consumption), live stock updates immediately
5. [X] Reject -> audit-logged; Store Keeper sees a status update

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site
2. Pick target site -> pick material -> see live availability
3. Add to cart, repeat
4. If cart > 5 items: an automatic WhatsApp escalation to the target HOD will fire on submit. Document why in Justification.
5. Click Submit All Requests to Admin

6.

Watch the [OK] My Requests tab - when status flips to "approved", you can mark it as fulfilled once received

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests
2. Review the request table
3. [OK] Approve -> physical handoff begins
4. [X] Reject -> use when the request is unjustified or your stock is too tight

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests
2. Create New PR
3. Type PR Number, pick Material (auto-fills SAP, MatCode, UOM), Qty, Supplier, Est. Cost SAR, Notes
4. Click Create PR Draft

Option B: PDF upload

1. Upload PR PDF
2. Drop the PDF in
3. Click Process Upload
4. System auto-extracts PR number + items, creates pr_master rows

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests
2. Select open PR in the email dropdown
3. Draft Outlook Email -> Mail/Outlook opens with HTML table pre-filled
4. Review/send
5. Also: Download PR PDF for your records

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds (cache TTL). Refresh the page. If it still doesn't, check audit log for the APPROVE_ADJUSTMENT entry to confirm the synthetic ledger row was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has

requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in receipts table, not pending_receipts.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is mailto: (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format GI-XXXXXXX. Items not in your inventory's Material_Code field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> Audit Logs by action FEFO_OVERRIDE. Or query the consumption table for FEFO_Override IS NOT NULL.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

render_brand_header("Reports & Analytics"). Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first, with	Answering "what is on our shelves worth?"
Shelf-Life / Expiry	Items approaching expiry, critical and warning	The weekly walk-round to pull short-dated stock

Report	What it shows	Typical use
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first,	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Calls `_run_report(type, from, to, site_id)` -> returns (DataFrame, summary_dict). Loading state shows shimmer.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	"<Type> - <site>" + "<from> -> <to>"
v Download <fmt> button	Encodes via <code>generate_report_pdf/excel/csv</code> and triggers <code>st.download_button</code>
Summary cards	Auto-generated from the summary dict keys
Bar chart	For Burn Rate and Daily reports
Preview table	Styled rows with status-color cells

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

Report icon + label

- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

If AI_ENABLED and Ollama is running with llama3.1:8b:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 fixed SQL probes + LLM commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	LLM-reported confidence
Right-side metric callout	Headline number with sub-label
Body paragraph	LLM-generated explanation grounded in the SQL probe result
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 fixed probes:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If Ollama is not running: a graceful fallback explains setup steps.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
- Re-email
- Delete (audit-logged)

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned} - \text{computed live in v_site_stock view.}$

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned /
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	read_at IS NULL (unread) -> timestamp (read)
delivery_reminders_sent	T-2/T-1/T-0 idempotency log	(terminal - UNIQUE constraint)

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	Substring RL-, RUBBER LINING, RUBBER-LINING in Material_Code OR Description
BL	Brick Lining	Substring BL-, BRICK LINING, BRICK-LINING, BRICK MATERIAL
NULL	Neither family	Default

Logic in the system. NEVER a combo string - RL takes precedence if both tokens are present.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13
